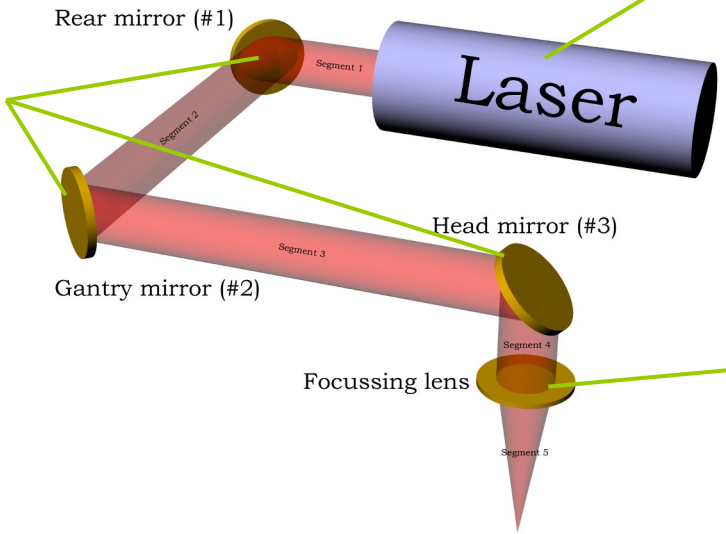
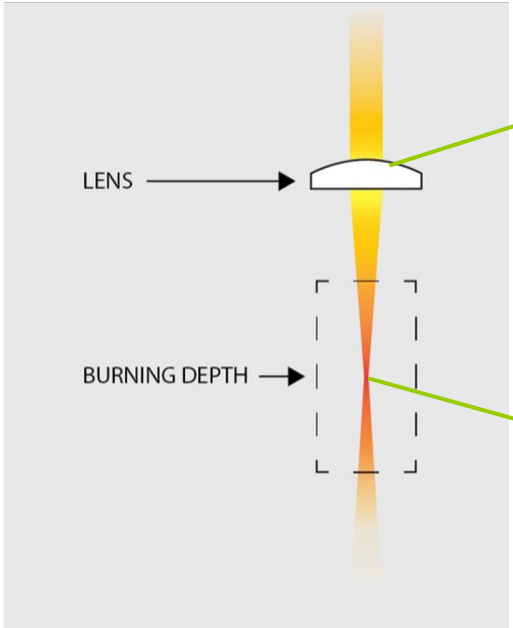
Lasercutting workshop

Die Laserquelle
erzeugt
einen Lichtstrahl ...



…, der durch Spiegel auf
den Schneidkopf geleitet ...

... und schließlich durch
eine Linse fokussiert wird.



Fokus-Linse ...

... fokussiert den Laser nach unten ...

... zu einem dünnen und konzentrierten Strahl

schneiden



gravieren



gravieren & markieren

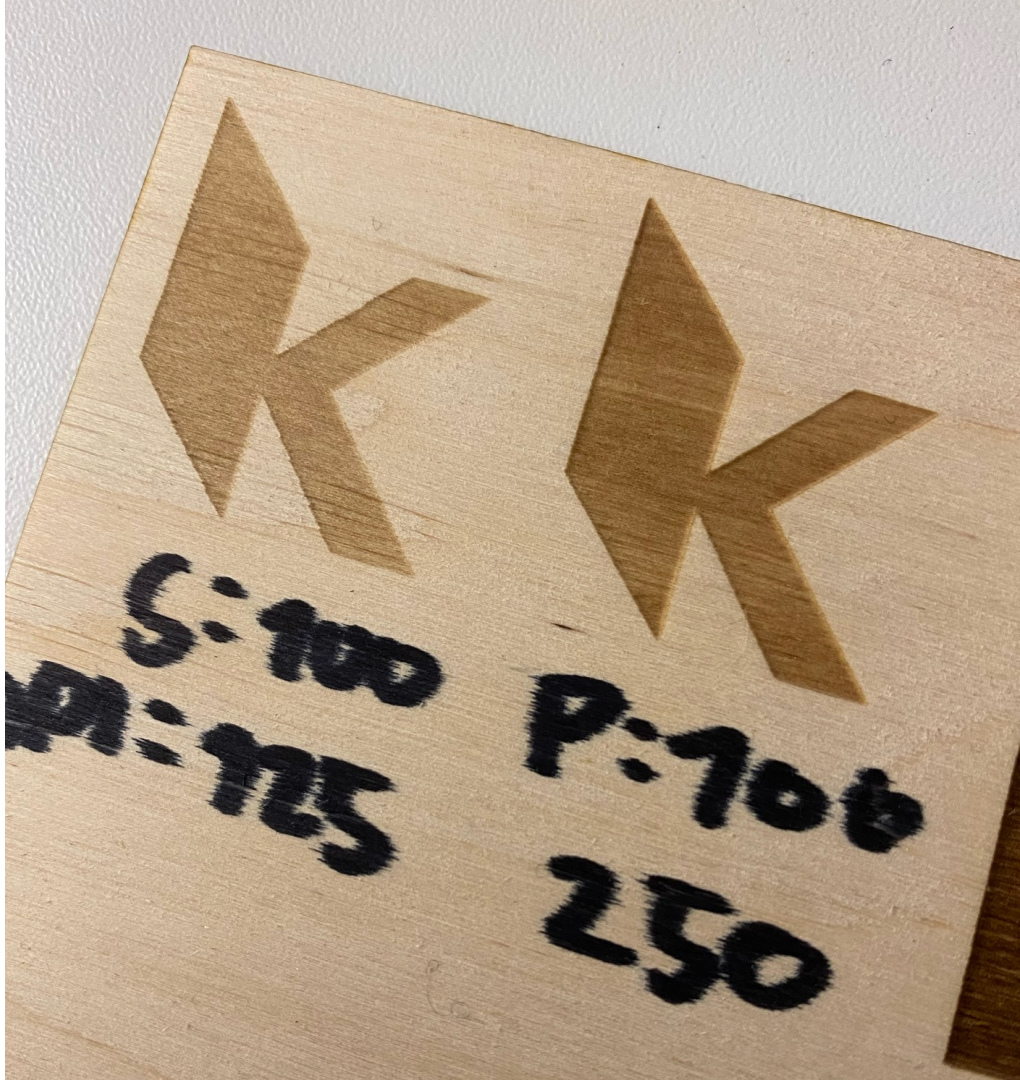


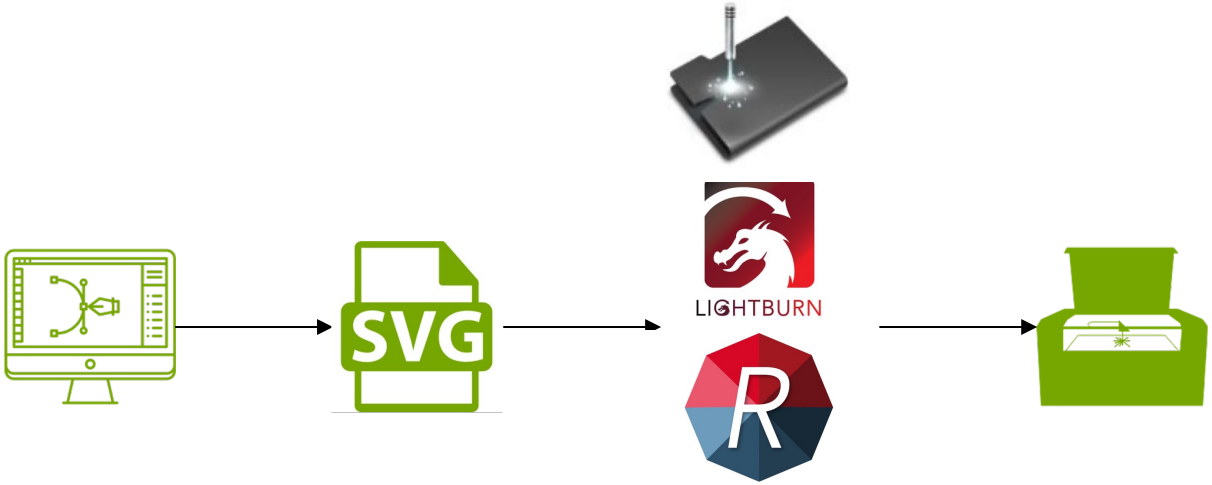
markieren



Reducing DPI can speed up jobs sigbnificantly.

full speed + full power allows to reduce DPI to 125 for light engraving and 250 for darker engraving. DPI has a linear relation to cutting time. Half the DPI half the time.





Erstellen einer Vektordatei

Exportformat .svg

+ .ai .pdf .dxf

Einrichten mit CNC-Software

Schnitt- und Gravur-Prozess

Luftunterstützung und Vacuair müssen immer eingeschaltet sein.

Bleibe während des Schnitt- und Gravurvorgangs bei der Maschine und überprüfe regelmäßig, ob alles in Ordnung ist.

STOPP bei konstanter Flammenbildung



Im Falle eines Vorfalls wende dich bitte an einen Werkstattleiter.

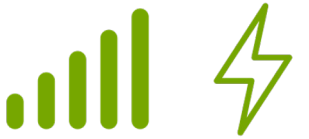


Lasse den gesamten Rauch aus dem Schneideraum absaugen bevor du den Deckel öffnest.

Power: **0,1–100 %**

Geschwindigkeit: **0,1–100 %**

Frequenz **0-5000hz**





Laser folgt Vektoren mit hoher Leistung und mittlerer Geschwindigkeit

→ Durchtrennen des Materials

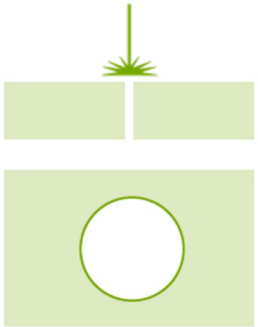
High Power



Medium Speed







.

Laser folgt Vektoren mit mittlerer Leistung und hoher Geschwindigkeit

→ Anschnitt der Oberfläche

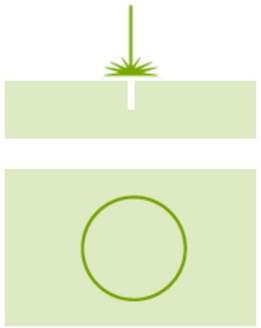
Medium Power



High Speed

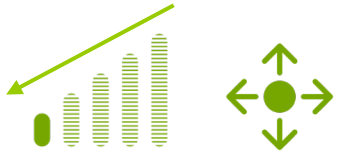


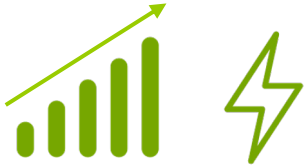




Geschwindigkeit verringern und/oder
die Leistung erhöhen.







Anzahl der Laserpulse pro Sekunde von 1 bis 5 000 Hz

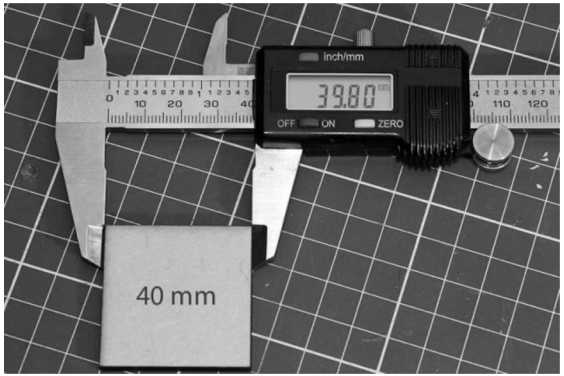


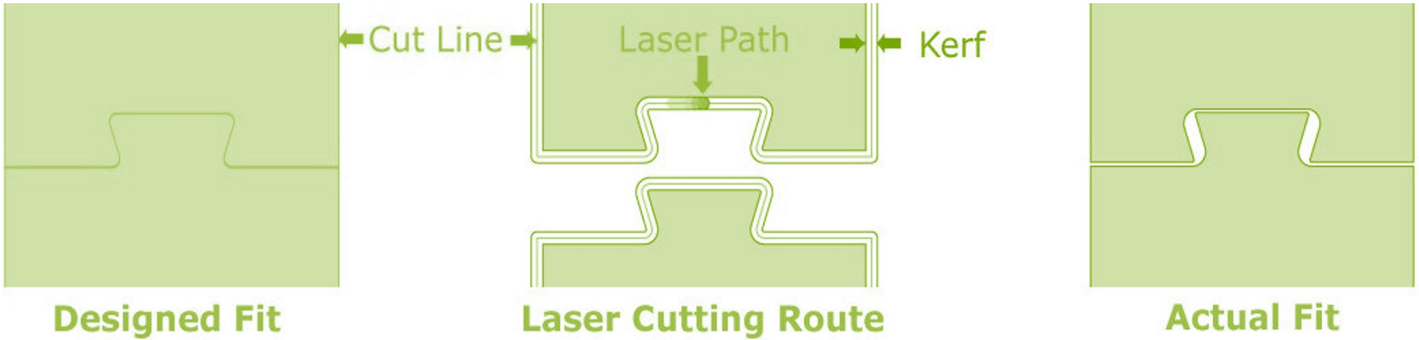
Verbrennende Materialien (Holz, Papier):
Niedrige Frequenz, 500 Hz

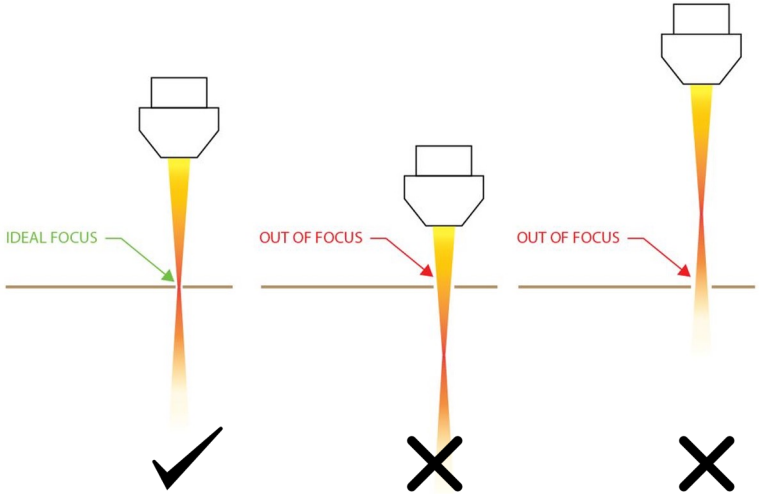


Schmelzende Materialien (Kunststoffe):
Hohe Frequenz, 5000 Hz

Abhängig von Materialeigenschaften
ca. 0,10–0,30 mm





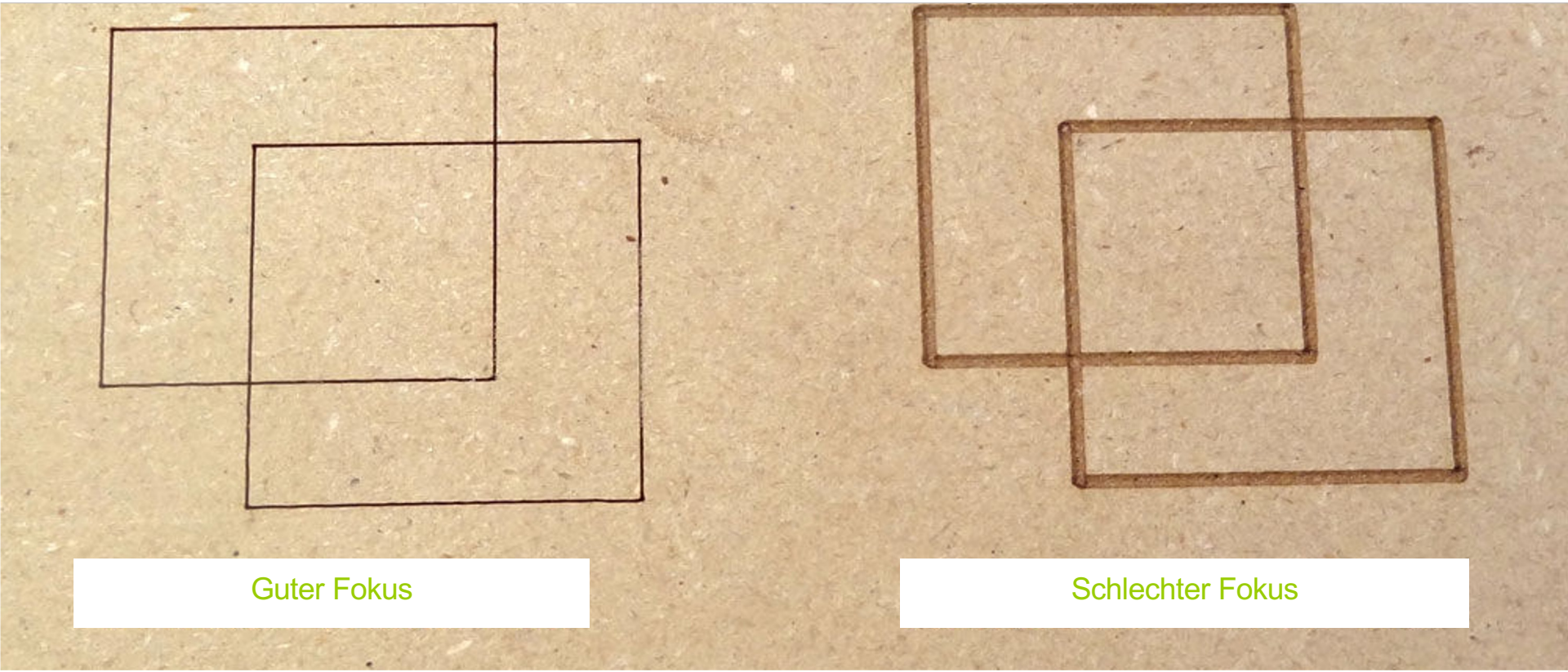


Idealer Fokus

dünnster Laserstrahlabschnitt
auf der Oberfläche des Materials

Defokussiert

Schneidet nicht durchs Material
dicke Schnittlinien

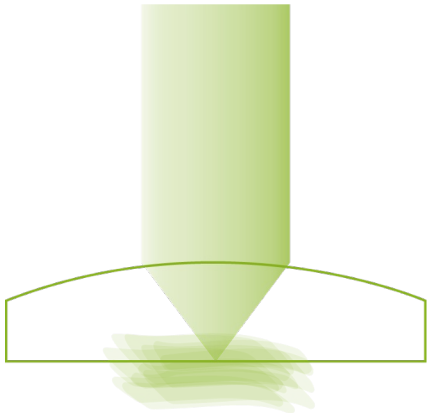


Guter Fokus

Schlechter Fokus

Beim Laserschneiden entstehen Rauch und Staub und die Linse wird **verschmutzt**.

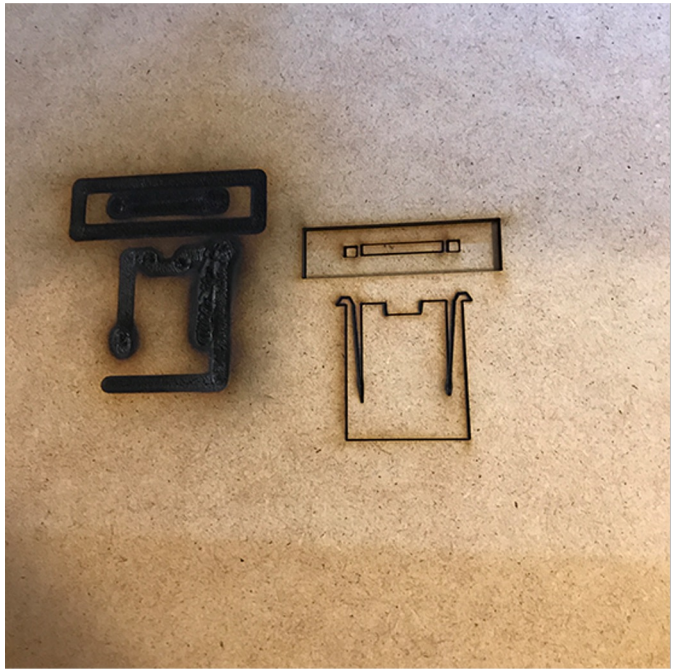
Bei zu starker Verschmutzung fokussiert der Laserstrahl auf die Linse und verbrennt sie.



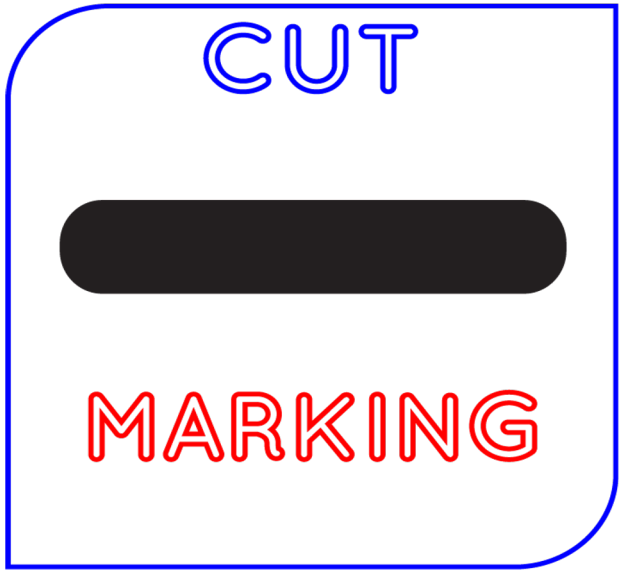
- Sieht schlecht fokussiert aus
- Flammen und Rauch
- Schneidet nicht durch
- Dicke und verschmutzte Schnittlinien

Sofort anhalten

Frage die Werkstattleitung nach einer Linsenreinigung.



Beginne immer mit einem Test in der Ecke deines Materials,
um die richtigen Einstellungen zu finden.



Zu dunkel oder verbrannt:

Leistung verringern
Geschwindigkeit erhöhen

Schneidet Material nicht vollständig durch:

Leistung erhöhen
Geschwindigkeit verringern

Erzeugen toxische Gase



Beschädigen die Maschine



Entflammungsgefahr





Verbotene Materialien

brennbare Stoffe,
z.B. Schaumstoffe fangen leicht Feuer
und sind i.d.R. verboten.

Metalle

Reflektierende Oberflächen

Materialien, deren Bestandsstoffe nicht zu ermitteln sind

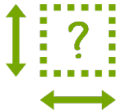
Vinyl
Polyvinyl Chloride (PVC)
Polyvinyl Butyral (PVB)
Polytetrafluorethylene PTFE (Teflon)
Neoprene (CR)
Kevlar, Nomex (PPTA, PMPI)
Polystyrene/ Polystyrol (PS)
Polycarbonate (PC)
Polycarbonate (PC)
Polyurethane (PUR)
Polyethylene (PE)
FPM / FKM (Viton®, Tecnoflon®, Fluorel®, Dai-el®)
Bakelite
Synthetic leather
Acrylonitrile butadiene styrene (ABS) Styrene

Caoutchouc
Latex

Beryllium oxide
Glass Fiber
Carbon or Carbon Fiber
Printed Circuit Board (PCB)

Farbmodus
RGB

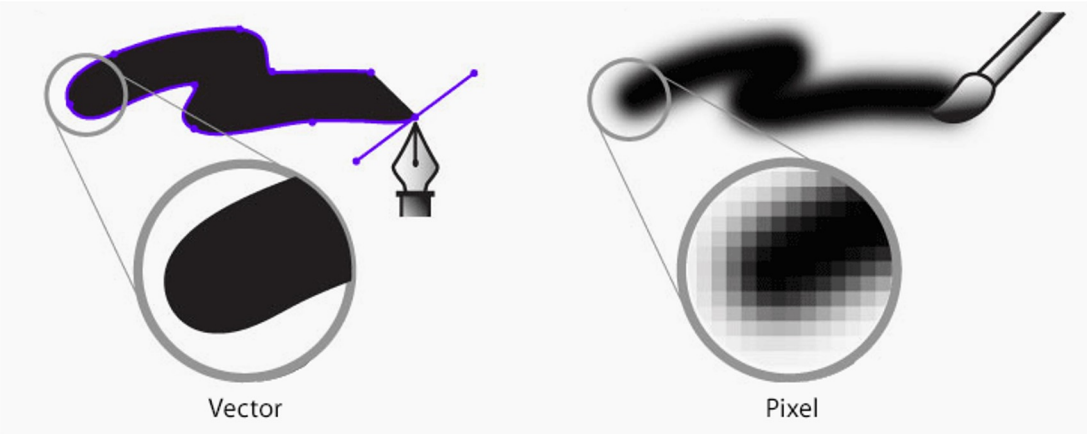




Abmessungen

Illustrator
Corel Draw
Autocad
Inkscape (free)
GravitDesigner (free)





Schneiden, Markieren und Gravieren möglich

Nur Gravieren möglich

Graviert jeden einzelnen Punkt einer Fläche

→ Flächiges Abtragen des Materials

Medium Power



High Speed













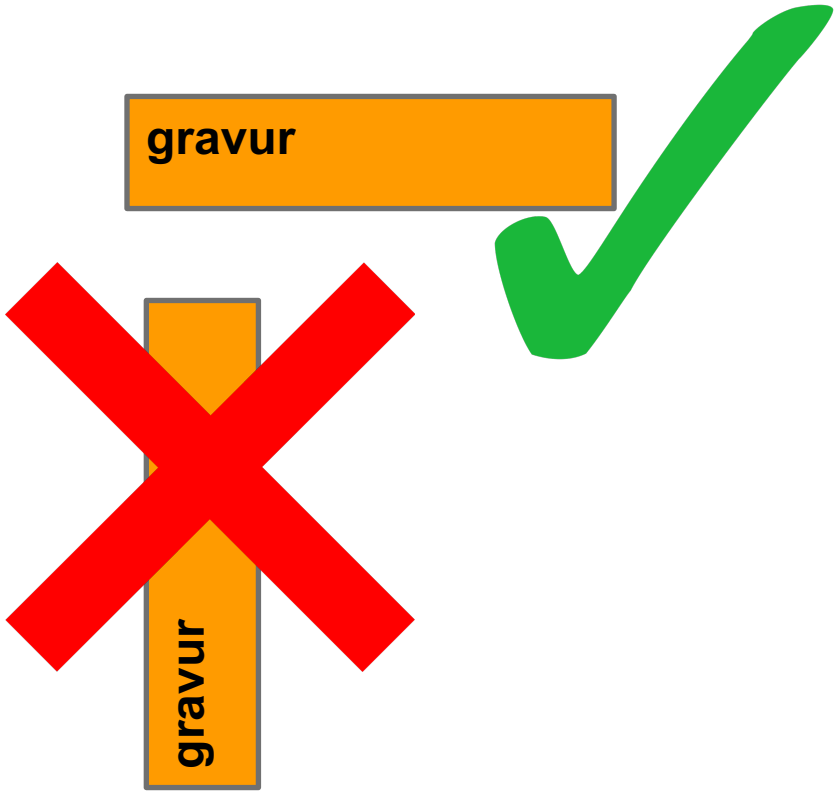


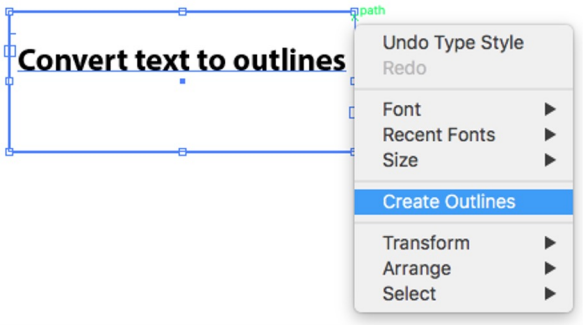
höhere Auflösung

längere Gravurzeit

höherer Detailgrad

- Gravuren entlang X-Achse anordnen





Convert text to outlines

Convert text to outlines

Bridges recommended for text cutting out

metroplex laser

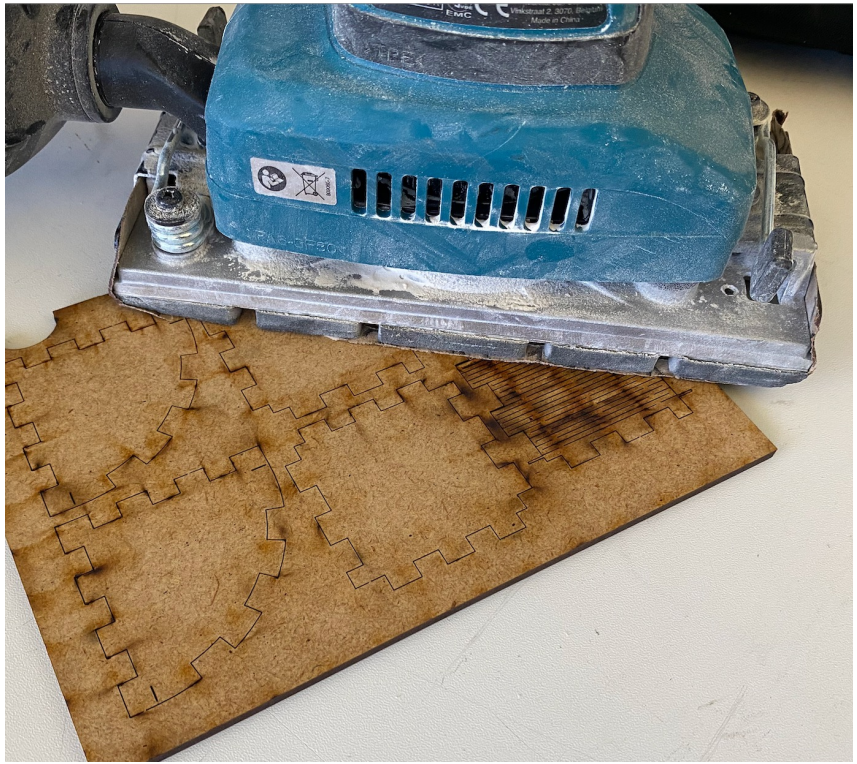
or use laser cutting fonts

preventing
burning
(David)

Prevent burning:

- Clean lens
- Focus correctly
- Make sure Air Assist is on
- Choose a setting that cuts through the material but is not overpowered.
- To prevent burnmarks from the metal grid, lifting the pieces is an option
- Potentially tape the material prior to cutting
- Apply water prior to cutting

Remove burn



Sand the whole plate while pieces are still in place.
It‘s quick and allows to remove heavy burn.

Sanding burn at
any later stage is harder.

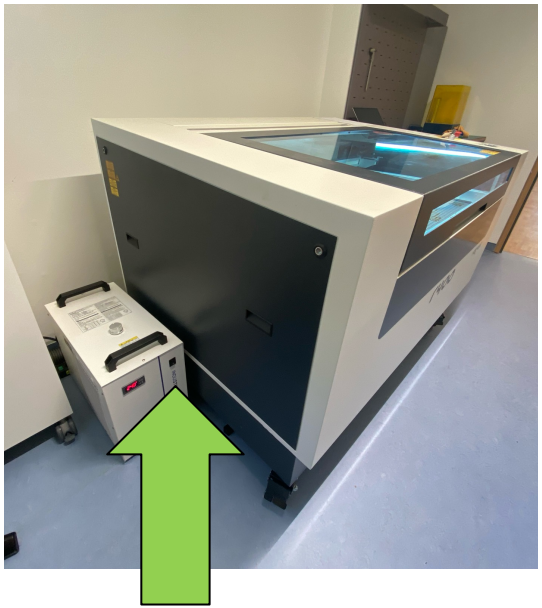
Remove burn on engravings



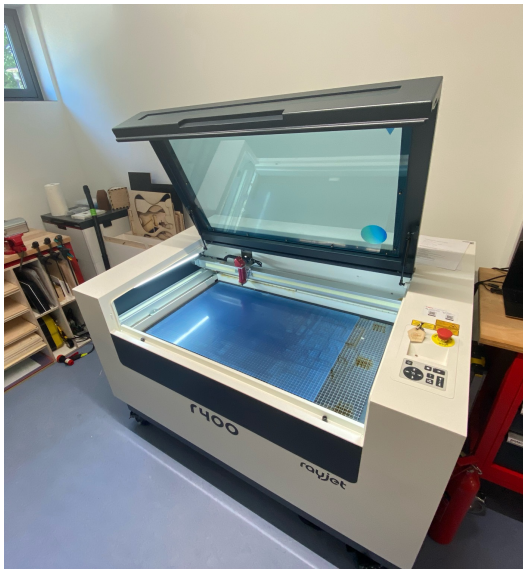
Use a soft wire brush to carefully remove burn on engravings

Starting the laser

Starting the laser



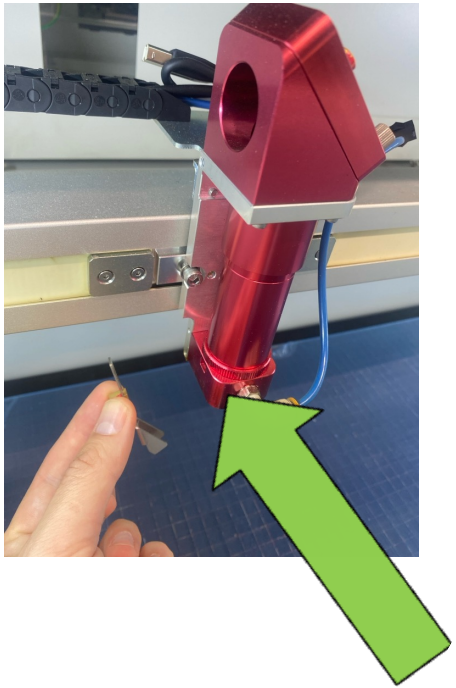




1) Switch on water cooling
2) Turn on laser with key
3) Open & Close lid and wait for homing
   (laser will approach endstops on all axis)

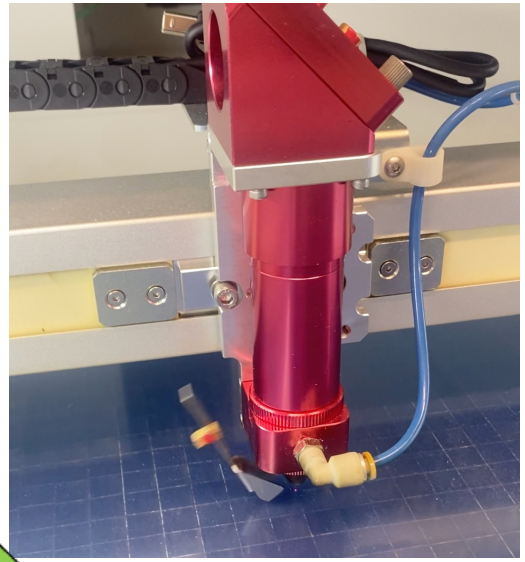
Manual focus:

Attach focus gauge



Move Z-axis up until gauge falls down

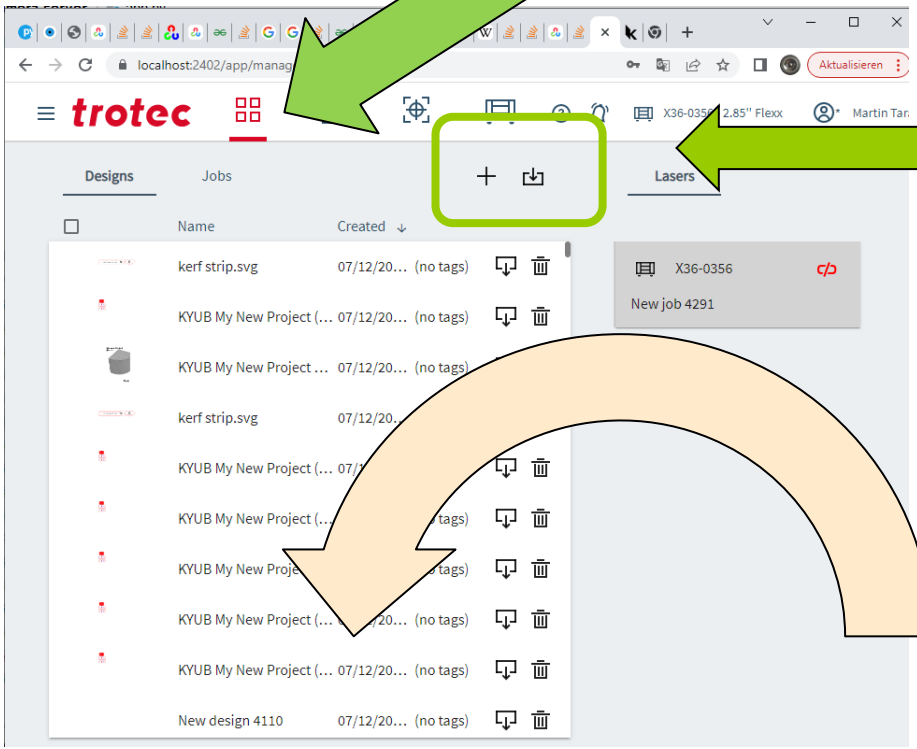




Using ruby software

Working with Ruby

Import files on first tab

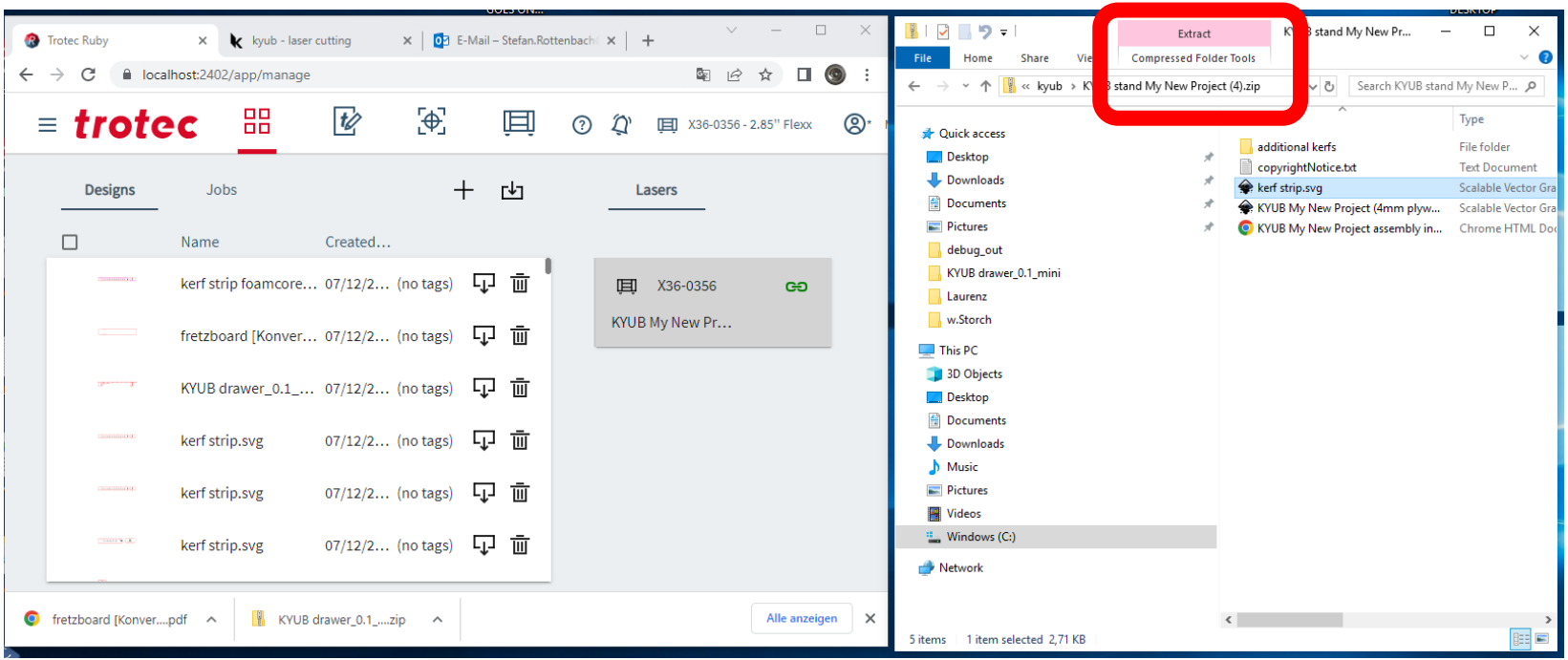


By file dialog, or



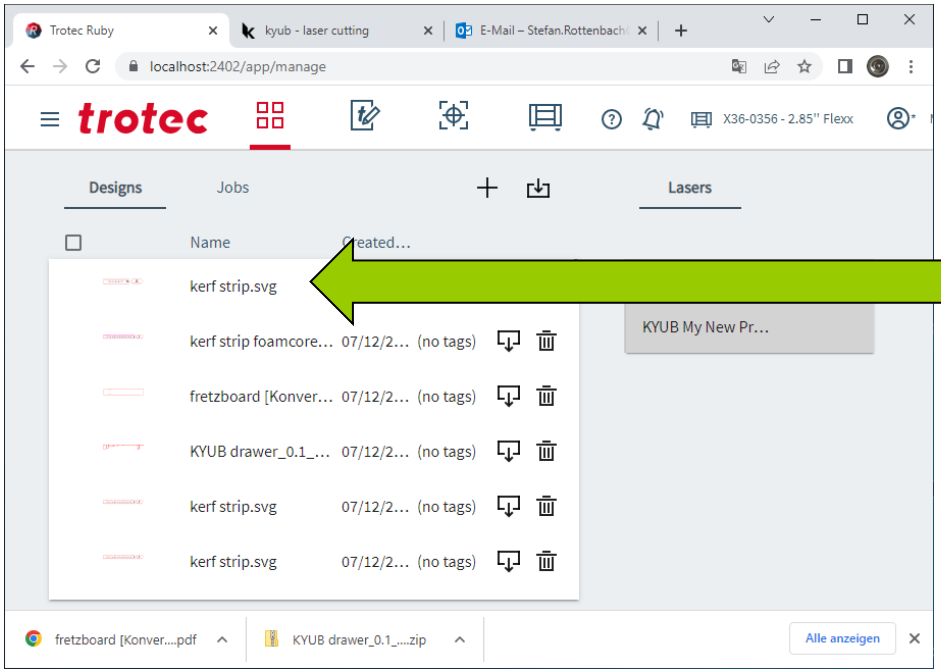
Drag and drop

Working with Ruby



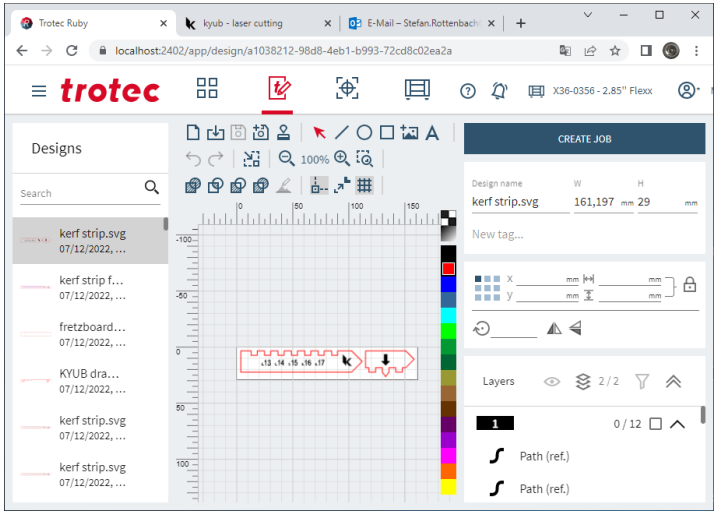
Dragging from a doubleclicked KYUB export zip file wont work.
Extract files first in order to drag them.

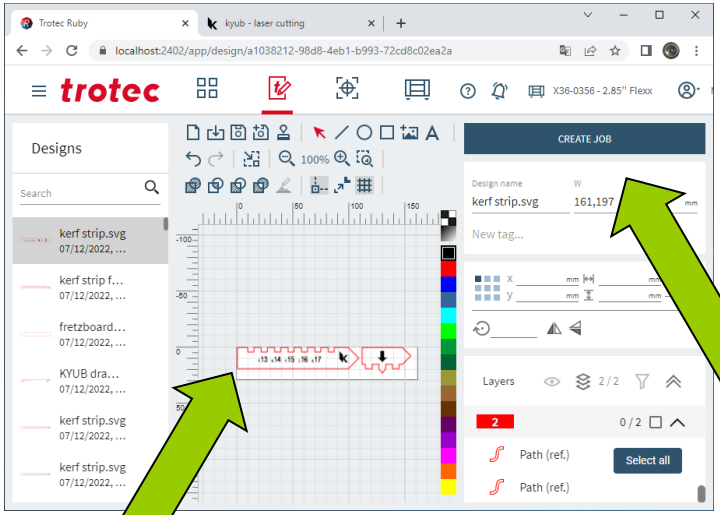
Working with Ruby



Doubleclick file to move to the next tab

Working with Ruby

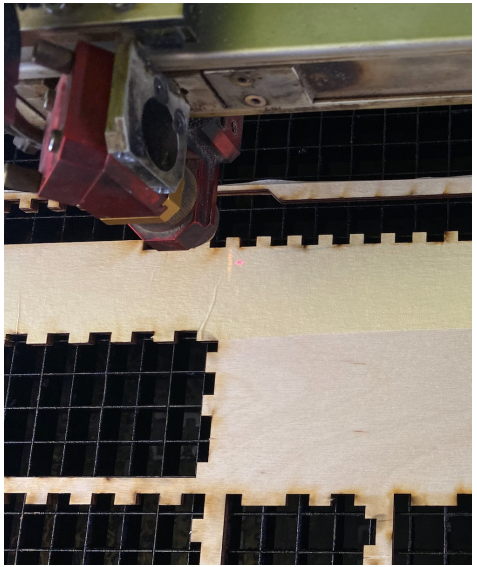


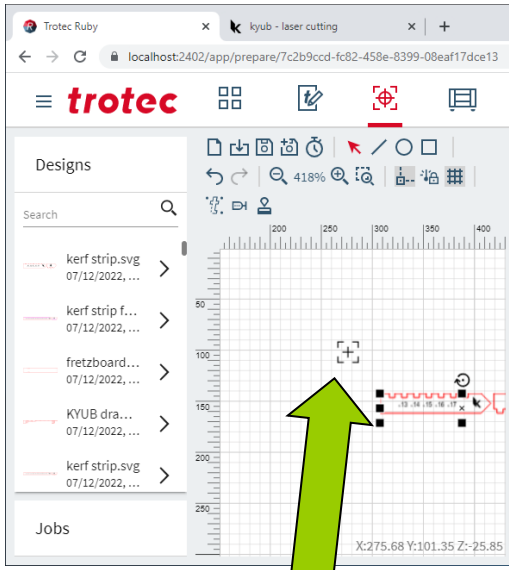


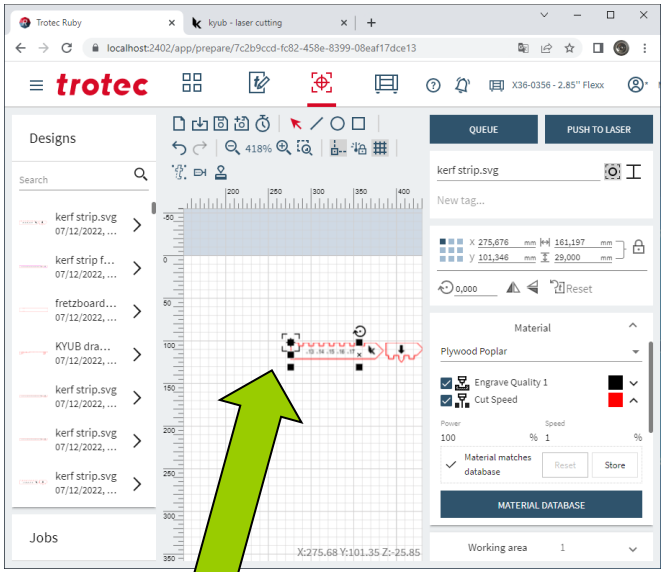
For precise alignment it‘s usefull
to move the file to the artboard boundary

After arranging press
„create job“

Working with Ruby





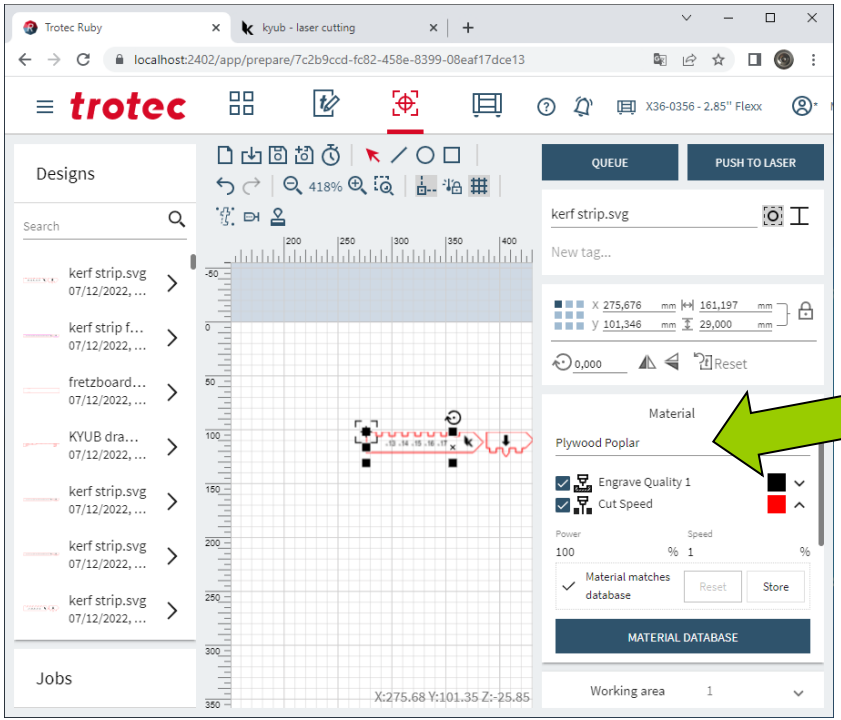


Move laser (red dot) to desired cutting area

Crosshair represents position of laser

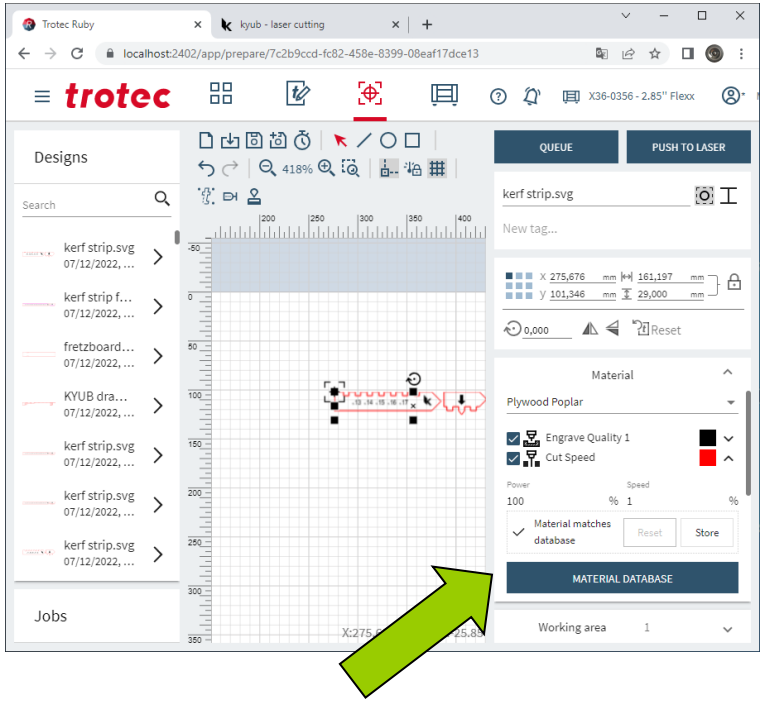
Align file with crosshair (magnetic snap)

Working with Ruby

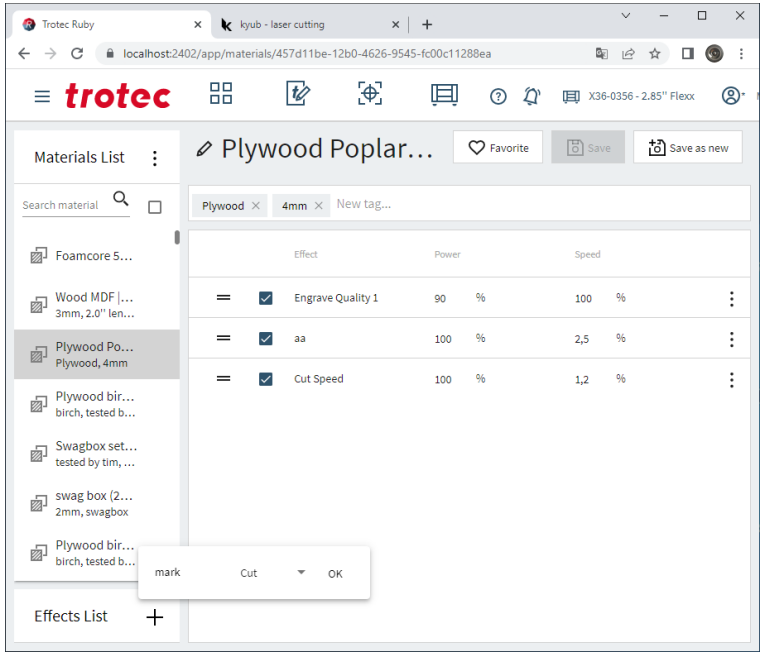


Select material here,
edit settings if needed

Working with Ruby

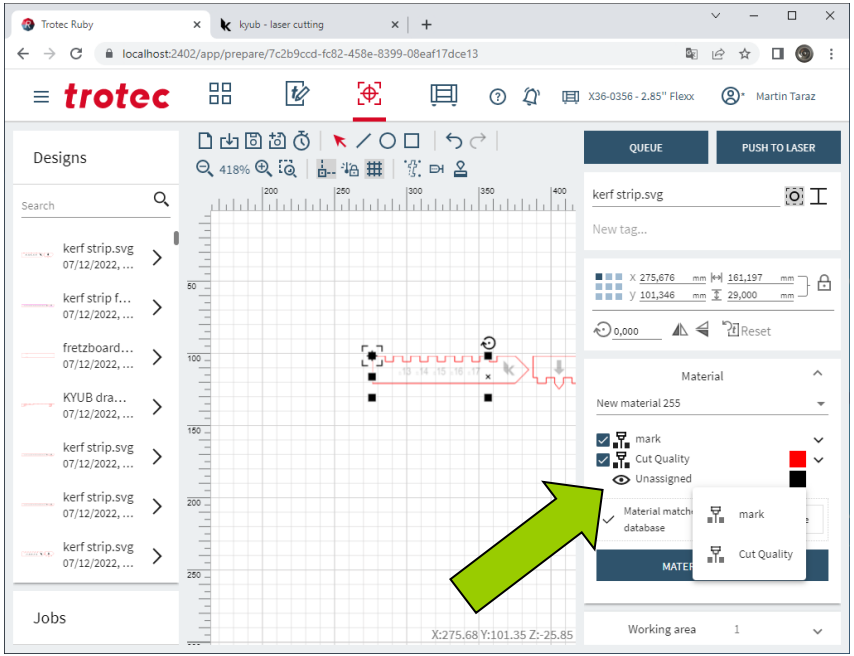


For all editing options jump to „material database“

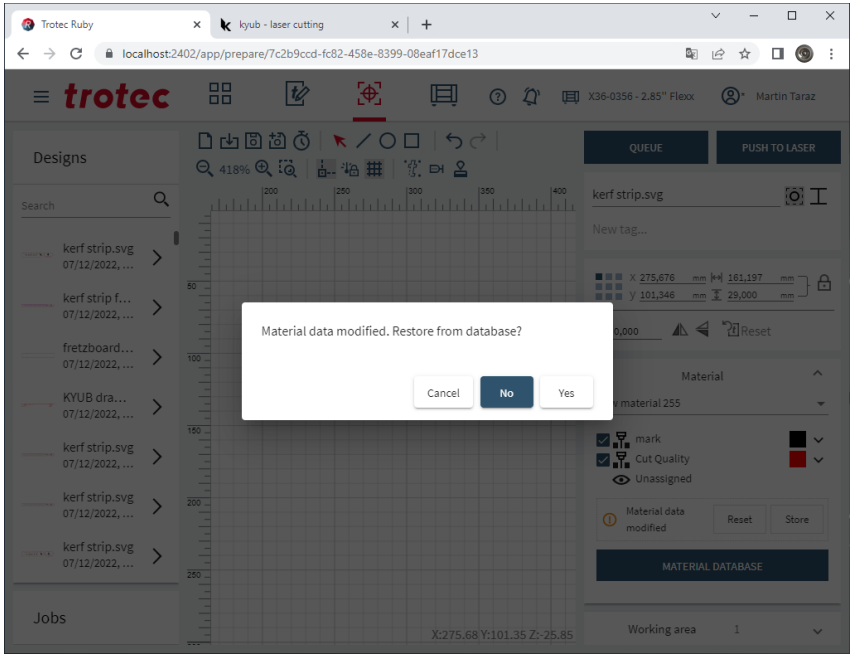


Here you can save new materials and rename them

Working with Ruby

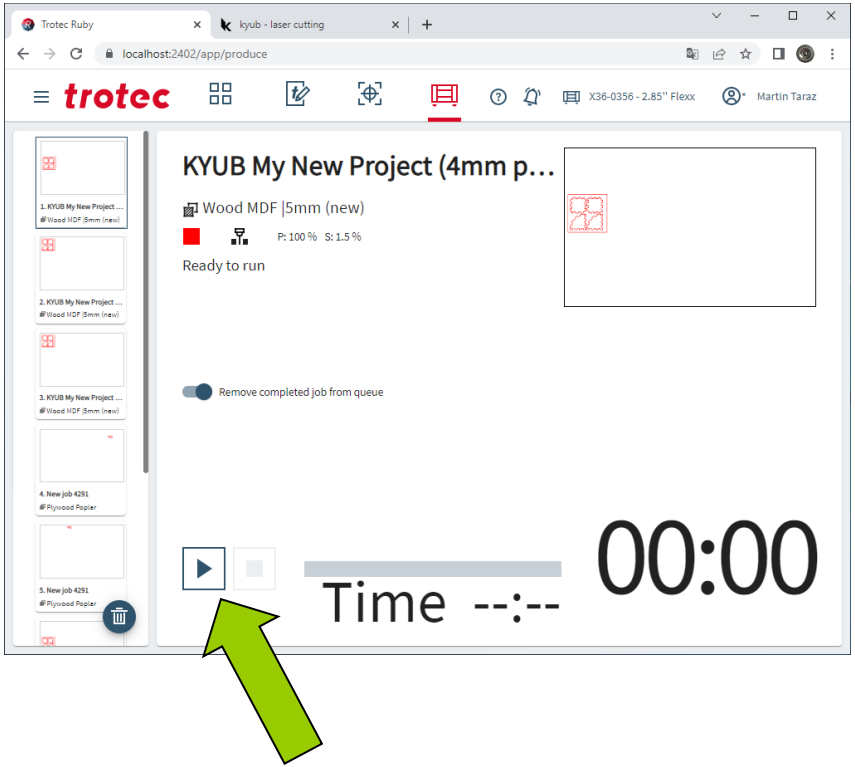


Assign colors to the predfined process here (right click on color)



If you made changes do not restore the defaults from datebase

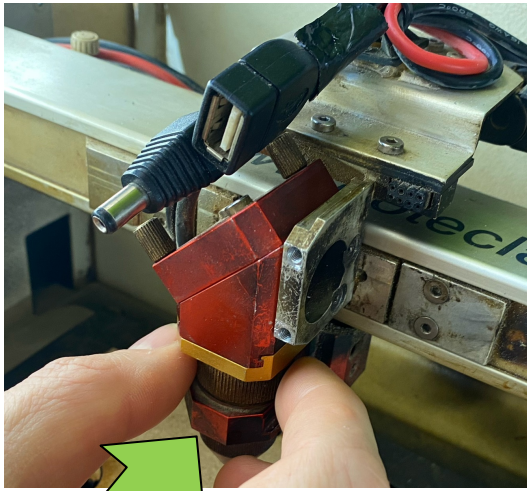
Working with Ruby



You reached the final tab,
Hit play to cut!

Laser maintenance

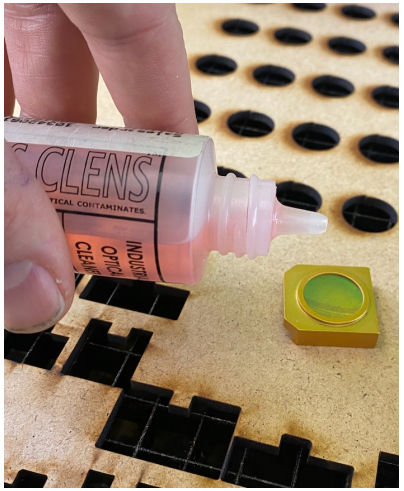
Before cutting: Check lens & clean if required

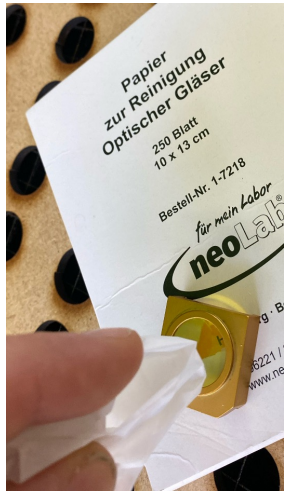


1) Unscrew below lens,
2) Remove lens and check



3) it‘s dirty



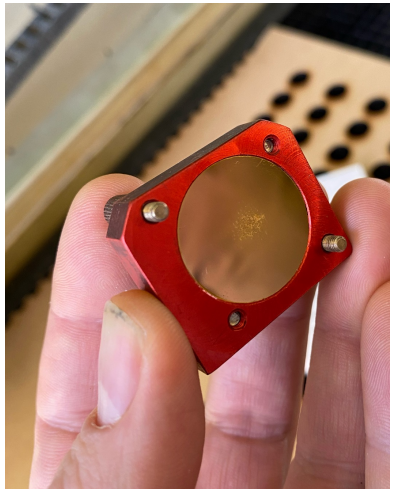


4) Apply 1 drop of cleaning fluid
5) Gently wipe with lens cleaning paper
6) insert lens and screw **tightly**

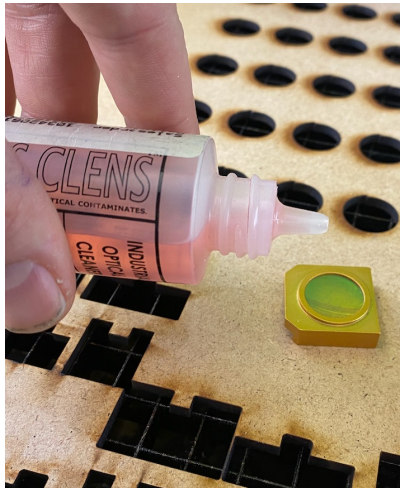
Before cutting: Check mirror & clean if required

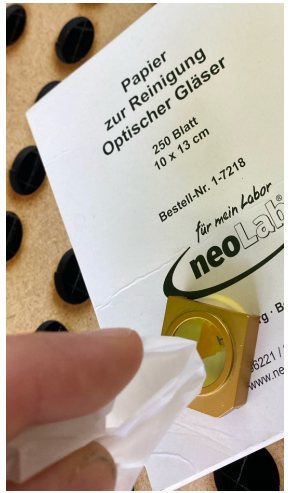


1) Unscrew mirror
2) Remove mirror and check



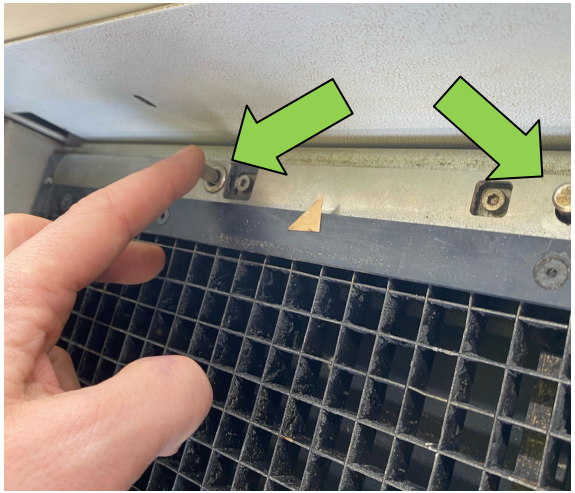
3) it‘s dirty





4) Apply 1 drop of cleaning fluid
5) Gently wipe with lens cleaning paper

Clean work table / remove grid



1) Move x-carriage to top
2) Click locking bolts to open
3) Remove **carefully**,
   do **not** hit the lens unit



4) Clean table with handbrush or vacuum.
It‘s important to do it regularly because leftovers can easily burn.